Temporal Prediction Produces Viewpoint-Consistent, Human-Aligned Vision in DNNs

Anonymous Author(s)

In submission to the *9th Conference on Cognitive Computational Neuroscience* (CCN 2026).

Abstract

A long tradition in computational neuroscience proposes that the temporal structure of visual experience shapes how biological systems learn to see. Deep neural networks (DNNs), typically trained on static labeled images, lack this inductive pressure. Could the learning regime explain their growing misalignment with human vision? We fine-tuned large pretrained vision transformers on naturalistic videos of real-world objects, systematically comparing self-supervised objectives that range from reconstructing the current frame to interpolating between frames to predicting the next frame. Next-frame prediction in pixel space produced the largest gain, raising alignment with humans to near the human ceiling while maintaining classification accuracy and near-human-level viewpoint stability. These results offer a step toward identifying the learning principles that wire up biological vision.

Keywords: Computer Vision; Human Alignment; Self-Supervised Learning; Biological Vision; Predictive Learning

Introduction

Deep neural networks (DNNs) now surpass human performance on many visual tasks, yet their ability to predict human behavior and neural activity has plateaued and, in some cases, worsened as accuracy has increased (Linsley et al., 2025). A key reason is that DNNs and humans rely on different visual features and strategies (Geirhos et al., 2022). One response is to directly supervise DNN feature maps with human psychophysics data (Fel et al., 2022), but this sidesteps a more fundamental question: what learning principles give rise to human-like vision in the first place? A long tradition in computational neuroscience holds that temporal structure in visual experience is central to learning invariant representations.

Földiák's trace rule showed how temporal continuity could teach viewpoint invariance (Földiák, 1991), and slow feature analysis formalized the broader principle that useful visual features are those that change slowly over time (Wiskott & Sejnowski, 2002). Recent work with deep networks supports this view: self-supervised objectives produce ventral-stream-like representations without category labels (Zhuang et al., 2021), predictive learning over video gives rise to dorsal/ventral specialization (Bakhtiari et al., 2021), and models trained on egocentric child video develop robust object representations (Lake et al., 2017; Orhan et al., 2024). Yet it remains unclear which aspects of temporal learning matter most: the temporal structure itself, the nature of the prediction task, or the space in which learning occurs.

We address this by fine-tuning vision transformers (Dosovitskiy et al., 2021) on naturalistic real-world video, systematically varying self-supervised objectives along two axes: temporal task (reconstruction, interpolation, or prediction) and target space (pixel or latent). We evaluate each model on classification accuracy and alignment with human visual strategies, measured via human psychophysics and viewpoint stability. Pixel Prediction (next-frame prediction in pixel space) produces the largest gain, increasing alignment from 0.017 to 0.63 (human ceiling: 0.71) while maintaining classification accuracy, with near-human-level viewpoint stability.

Approach and Method

Data generation. We used 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) to generate naturalistic video with two properties not available in standard datasets: smooth, controlled viewpoint trajectories for training, and ground-truth 3D geometry for measuring representation stability across viewing angles. We fit 27,823 3DGS models to Common Objects in 3D (Co3D) (Reizenstein et al., 2021), which contains videos of real-world objects from 50 MS-COCO categories (Lin et al., 2014), and rendered 50 frames along a smooth trajectory per object. Models are trained on sequences of 8 frames sampled at a step size of 4.

Human feature importance maps were collected using ClickMe (Linsley et al., 2019), a large-scale online psychophysics platform where participants identify the image regions most relevant to recognizing an object's category. Each importance map is the average across multiple participants.

Model training. To understand how well DNNs align with humans on our dataset, we trained linear probes on 495 DNNs from TIMM (Wightman, 2019) and measured their classification accuracies and their alignment with humans. To explore the space of training routines, we fine-tuned an ImageNet-21K (Ridnik et al., 2021) pretrained BEiT-V2 Large (Peng et al., 2022) from TIMM with seven self-supervised objectives, along with supervised classification as control. The starting model was selected as a representative SOTA DNN due to its high classification accuracy on our dataset. The seven self-supervised objectives vary along two axes: the temporal task and the target space. The temporal task is one of *reconstruction* (recover the current frames from full or masked input), *interpolation* (fill in intermediate frames in a sequence), or *prediction* (generate the next frame from preceding context). Reconstruction includes MAE (He et al., 2021), VideoMAE (Tong et al., 2022), and autoencoder (Hinton & Salakhutdinov, 2006) variants. For interpolation and prediction, we use teacher forcing (Williams & Zipser, 1989) to leverage the full temporal structure. Each temporal objective is implemented in both pixel space and latent space, the latter following a DINO-style (Caron et al., 2021) design. During self-supervised fine-tuning, we also regularize the model using embeddings from the original model to prevent catastrophic forget-

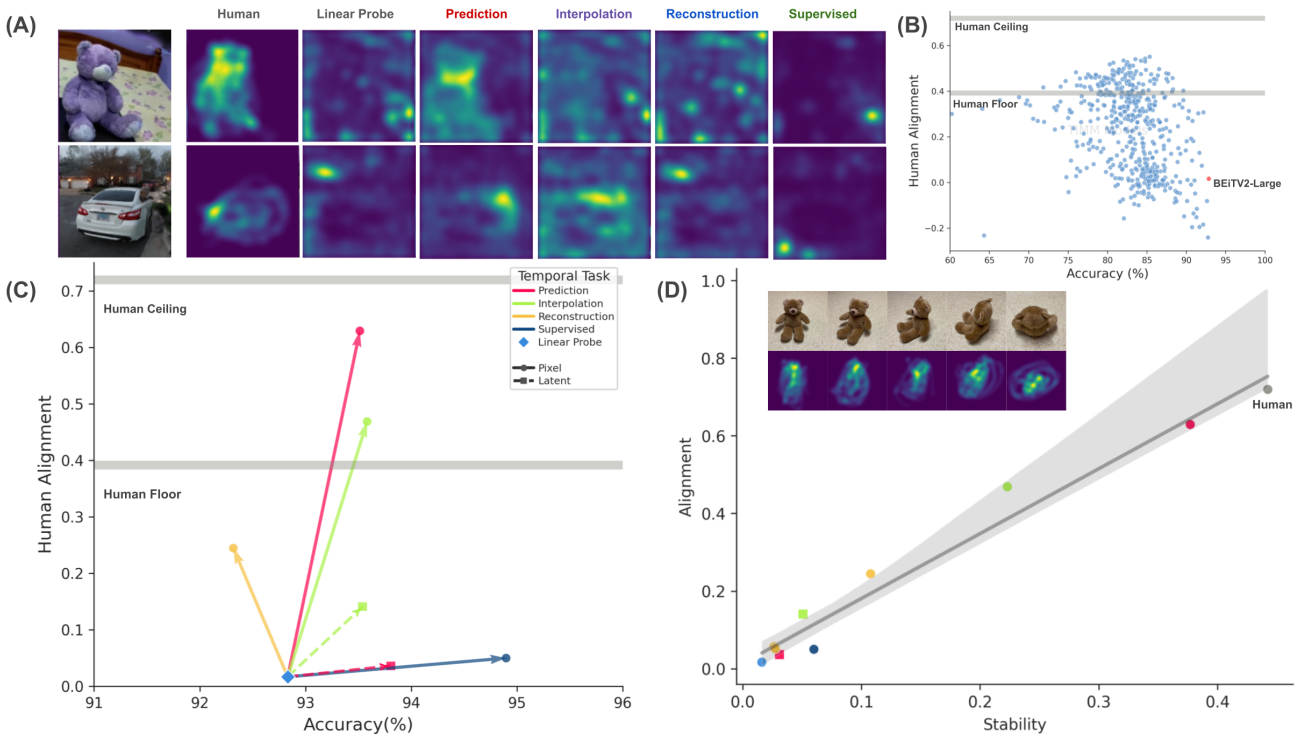

Figure 1: **Temporal Prediction aligns DNNs with human vision. (A)** Feature importance maps from humans and models fine-tuned with different temporal tasks. Prediction shifts model importance toward human-like feature strategies. **(B)** Alignment (correlation between human and model importance maps) vs. classification accuracy of 495 pretrained DNNs from TIMM. **(C)** Effects of different temporal tasks on BeitV2-Large (Peng et al., 2022). Pixel Prediction achieves the highest alignment without sacrificing accuracy. **(C)** Alignment vs. stability of humans and fine-tuned models. Pixel Prediction produces near-human-level stability. Stability is measured by projecting importance maps across viewpoints and computing correlations at each target angle. Embedded images are example images and human importance maps across viewpoints.

ting of learned visual features. For each task, we selected the variation with the greatest effect on alignment to represent that task's effect.

Model Evaluation. Model importance maps are computed by training a linear classification probe on each model and differentiating the output score with respect to the input image. We then measure human alignment as the Spearman's rank correlation coefficient between the human and model importance maps.

We evaluated human and model representation stability by projecting importance maps from one angle of an object to other angles, and computing Spearman's rank correlation coefficients between the projected maps and the maps from each target angle. We take the average correlation across angles as the score. The projection is achieved using 3D reconstructions from 3D Gaussian Splatting and camera matrices estimated with COLMAP (Schönberger & Frahm, 2016; Schönberger et al., 2016).

Results

Among 495 pretrained DNNs, alignment with humans increases with classification accuracy for weaker models but decreases for stronger ones (Fig. 1B), consistent with prior reports of a divergence between DNN performance and biological plausibility. Self-supervised fine-tuning with temporal objectives breaks this trend. Pixel Prediction produces the largest alignment gain, raising correlation between human and model importance maps from 0.017 to 0.63 (human ceiling: 0.71) without loss of classification accuracy (Fig. 1C). Qualitatively, Pixel Prediction importance maps shift toward the foreground features highlighted in human maps (Fig. 1A). Latent-space variants of the same temporal objectives yield much smaller improvements, indicating that alignment depends on pixel-level optimization, not just temporal structure. Pixel Prediction also produces near-human-level representation stability across viewing angles (Fig. 1D), while other objectives degrade more steeply with angular distance.

References

Bakhtiari, S., Mineault, P., Lillicrap, T., Pack, C. C., & Richards, B. A. (2021). The functional specialization of visual cortex emerges from training parallel pathways with self-supervised predictive learning. *Advances in Neural Information Processing Systems*, *34*.

Caron, M., Touvron, H., Misra, I., Jégou, H., Mairal, J., Bojanowski, P., & Joulin, A. (2021). Emerging properties in self-supervised vision transformers. *Proceedings of the International Conference on Computer Vision (ICCV)*.

Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. *ICLR*.

Fel, T., Rodriguez Rodriguez, I. F., Linsley, D., & Serre, T. (2022). Harmonizing the object recognition strategies of deep neural networks with humans. *Advances in neural information processing systems*, *35*, 9432–9446.

Földiák, P. (1991). Learning invariance from transformation sequences. *Neural Computation*, *3*(2), 194–200.

Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2022). Imagenet-trained cnns are biased towards texture; increasing shape bias improves accuracy and robustness. https://arxiv.org/abs/1811.12231

He, K., Chen, X., Xie, S., Li, Y., Dollár, P., & Girshick, R. (2021). Masked autoencoders are scalable vision learners. https://arxiv.org/abs/2111.06377

Hinton, G. E., & Salakhutdinov, R. R. (2006). Reducing the dimensionality of data with neural networks. *Science*, *313*(5786), 504–507. https://doi.org/10.1126/science.1127647.

Kerbl, B., Kopanas, G., Leimkühler, T., & Drettakis, G. (2023). 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, *42*(4).

Lake, B. M., Ullman, T. D., Tenenbaum, J. B., & Gershman, S. J. (2017). Building machines that learn and think like people. *Behavioral and Brain Sciences*, *40*, e253. https://doi.org/10.1017/S0140525X16001837.

Lin, T.-Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Lawrence Zitnick, C., & Dollár, P. (2014). Microsoft COCO: Common objects in context.

Linsley, D., Feng, P., & Serre, T. (2025). Better artificial intelligence does not mean better models of biology. *Trends in Cognitive Sciences*. https://doi.org/https://doi.org/10.1016/j.tics.2025.11.016.

Linsley, D., Shiebler, D., Eberhardt, S., & Serre, T. (2019). What are the visual features underlying human versus machine vision? *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*.

Orhan, A. E., Wang, W., Wang, A. N., Ren, M., & Lake, B. M. (2024). Self-supervised learning of video representations from a child's perspective. *CogSci*.

Peng, Z., Dong, L., Bao, H., Ye, Q., & Wei, F. (2022). BEiT v2: Masked image modeling with vector-quantized visual tokenizers.

Reizenstein, J., Shapovalov, R., Henzler, P., Sbordone, L., Labatut, P., & Novotny, D. (2021). Common objects in 3d: Large-scale learning and evaluation of real-life 3d category reconstruction. *International Conference on Computer Vision*.

Ridnik, T., Ben-Baruch, E., Noy, A., & Zelnik-Manor, L. (2021). Imagenet-21k pretraining for the masses. https://arxiv.org/abs/2104.10972

Schönberger, J. L., & Frahm, J.-M. (2016). Structure-from-motion revisited. *Conference on Computer Vision and Pattern Recognition (CVPR)*.

Schönberger, J. L., Zheng, E., Pollefeys, M., & Frahm, J.-M. (2016). Pixelwise view selection for unstructured multi-view stereo. *European Conference on Computer Vision (ECCV)*.

Tong, Z., Song, Y., Wang, J., & Wang, L. (2022). Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. https://arxiv.org/abs/2203.12602

Wightman, R. (2019). Pytorch image models. https://doi.org/10.5281/zenodo.4414861.

Williams, R. J., & Zipser, D. (1989). A learning algorithm for continually running fully recurrent neural networks. *Neural Computation*, *1*, 270–280. https://api.semanticscholar.org/CorpusID:14711886

Wiskott, L., & Sejnowski, T. J. (2002). Slow feature analysis: Unsupervised learning of invariances. *Neural Computation*, *14*(4), 715–770.

Zhuang, C., Yan, S., Nayebi, A., Schrimpf, M., Frank, M. C., DiCarlo, J. J., & Yamins, D. L. K. (2021). Unsupervised neural network models of the ventral visual stream. *Proceedings of the National Academy of Sciences*, *118*(3), e2014196118. https://doi.org/10.1073/pnas.2014196118.